ISP SECURITY COMMAND CENTER

Full system context and seven-day demo plan

A reusable, self-hosted sales demonstration for internet service providers

Decision: build a local, synthetic-data demonstration around an unchanged Wazuh core, with a separate custom dashboard, local-model incident summaries, and controlled n8n workflows.

Purpose: show prospective customers how a tailored security operations experience could work. This is a demo, not a deployment for any named person or customer.

Planning status: proposed architecture and implementation plan. No completed integration, deployed system, detection performance, or production readiness is claimed.

Prepared 31 October 2026 | Assumptions and upstream references included

1. Product vision and boundaries

An ISP-facing command center can turn technical alerts into a practical view for a security operator: what needs attention, which simulated infrastructure is affected, what evidence exists, and what a human might investigate next. The sales proposition is a configurable integration and service built around established components, not a claim that the underlying Wazuh platform was created by us.

Demo goals

- Show a convincing alert-to-investigation-to-workflow story in 10-15 minutes, with no real subscriber, customer, or production data.
- Keep the Wazuh agent, manager, indexer, dashboard and detection logic upstream and unchanged. Custom software talks to supported interfaces; optional configuration and test rules are separate, versioned files.
- Operate on one isolated local machine, replay repeatable fictional ISP scenarios, recover quickly, and clearly identify synthetic content and AI-generated text.
- Separate reusable product components from future per-customer configuration, credentials, branding, and deployment decisions.

MVP scope

One local operator experience: overview, alert list and filters, incident detail with evidence, AI summary, analyst acknowledgement, and a simulated workflow run with visible outcome. Use a small fictional topology and three curated scenarios. A seeded-data mode is required; a real isolated Wazuh integration is a stretch goal for the same seven days.

Not included in the demo

Real customer integrations, real network access, multi-tenancy, production-grade availability, automatic blocking or remediation, compliance certification, breach guarantees, and a promise that every ISP device can be monitored. A customer deployment requires discovery, implementation and validation under a separate agreement.

Assumptions

One developer with basic Docker, Python or TypeScript, web UI, and security operations experience; one demo laptop or workstation; a small local model already downloaded when presenting offline. Seven days is an ambitious engineering estimate, not a guaranteed delivery date. Hardware, model choice and event sources need confirmation.

2. System context

The intended end state has a clean boundary between upstream security components and our product layer. For the first demonstration, synthetic replay can stand in for Wazuh alerts. Labels must always reveal which path is active.

Proposed data flow

Fictional events / isolated lab sources
1. Seeded event replay, or optional lab-only log generator and agent
2. Wazuh manager decodes and evaluates events; Wazuh indexer stores alerts
3. Read-only integration service queries the indexer and normalizes selected fields
4. Custom dashboard shows incidents, evidence and operator actions
5. Local AI service generates a bounded, evidence-linked summary on request
6. Approved workflow request goes to n8n; n8n writes a simulated result back

For the minimum viable path, step 1 feeds the integration service from a local fixture file and steps 2-3 are explicitly marked simulated. When the isolated Wazuh path is enabled, show the original alert and its source identifier. The existing upstream Wazuh dashboard remains separately available for authorized technical investigation.

Actors and trust boundaries

- Presenter / analyst: views synthetic incidents, requests a summary, acknowledges, and runs a harmless workflow. The presenter verifies content before describing it.
- Integration service: the only custom component holding upstream credentials; validates and normalizes alerts, restricts reads, and mediates AI and workflow requests. The browser never receives Wazuh credentials.
- Local model: receives a limited incident excerpt as untrusted data; it cannot call tools, execute commands or choose actions. Model output is advisory and can be wrong.
- n8n: receives validated synthetic incident IDs and approved workflow type; demo workflows write only to a local log or simulated ticket queue. No external notification or destructive action is enabled.

Proposed technology

Docker Compose for repeatable local services; official Wazuh single-node Docker stack when resources allow; custom web frontend and small backend; local Ollama-compatible model API; n8n as a separately managed workflow service; local persistent storage for demo incident states and audit log. Exact framework, pinned versions, model and image digests are implementation decisions after a compatibility check.

3. Components and integration contracts

Wazuh and the adapter

Do not fork or modify core source. Use the upstream deployment as published, pin the chosen release and keep its notices. The manager generates alerts; the indexer stores searchable alert documents. The backend uses a least-privilege, read-only indexer account restricted to the required alert indices and fields, with TLS validation. The server API is a separate management interface; it is not assumed to be the alert-search interface. Preserve upstream timestamps, agent identity, rule ID and original event reference.

Adapter contract: list alerts with time window, severity and pagination; fetch one alert by source ID; map fields to the incident model; report connection freshness and errors. Handle field differences and unavailable indices explicitly. Never expose arbitrary index queries to the browser. A snapshot or fixture adapter implements the same contract in simulation mode.

Normalized incident record

Minimum fields: incident_id, source_mode (synthetic or wazuh_lab), source_event_id, observed_at in UTC, severity, title, category, asset label, site, source/destination indicators if present, evidence excerpts, rule ID, status, and last_updated. Use documentation-only IP ranges for examples. Include an explicit synthetic flag and retain provenance; do not present seeded events as detections by Wazuh.

Local AI summary

The backend sends only the selected, size-limited incident fields to a local model endpoint such as Ollama POST /api/generate. Request a structured response: observed facts with evidence references, possible interpretation, uncertainty, and up to three analyst checks. Validate the schema, cap length and timeout, and display the model name plus generation time. Treat log text as data, never instructions. If the model is missing or fails, show a clearly labeled prewritten demo summary or an unavailable state; never silently call a cloud provider.

Workflow contract

The dashboard submits incident_id, workflow_type, request_id and synthetic=true to the backend. The backend checks an allowlist and posts to a protected n8n webhook. n8n validates the payload, deduplicates request_id, records a local simulated ticket or notification, and returns execution ID and outcome. Set timeouts and bounded retries; show pending, success or failure. The demo does not route to real email, messaging, tickets, firewalls or subscriber equipment.

4. What the dashboard should show

Overview

A visible DEMO / SYNTHETIC banner and source-mode indicator; counts by severity and status; a small time trend; and a short list of recent incidents. Counts must be computed from the loaded data, with the displayed time range and last refresh. Avoid invented ROI or coverage claims.

Incident queue and detail

Filter by severity, category, site, status and time. Each incident opens a detail panel showing source, event time, fictional asset, rule or fixture identifier, raw evidence excerpt, activity history and a link to the Wazuh event only when the lab integration actually supports it. Analyst can acknowledge and add a local note. Show storage and reset behavior to the presenter.

AI and workflow panels

The AI panel separates source facts, hypotheses, unknowns and suggested human checks. Display "AI-generated; verify against evidence" with model and timestamp. An explicit button triggers generation. A workflow button previews the intended action and then runs the local simulation, displaying request and execution status. Include a non-AI fallback so the live presentation does not depend on model latency.

Three synthetic scenarios

- Authentication abuse: repeated unsuccessful logins against a fictional management server, then a successful login. Show counts and timestamps; do not assert account compromise without evidence.
- Infrastructure configuration change: a fictional router configuration file changed outside a stated maintenance window. Explain that source and decoder support must be established in a real deployment.
- Malware or suspicious process: an isolated lab endpoint produces a simulated high-severity alert. Show the original evidence and a human validation checklist, not an automatic quarantine.

Fixture time offsets should move with each reset, so the dashboard looks current without misrepresenting event timestamps. Label all hostnames and entities fictional and avoid real customer logos.

Demo completion criteria

- On a clean machine, one documented start procedure opens the dashboard on loopback and loads three synthetic scenarios; reset returns to the same known state.
- Operator filters, opens evidence, acknowledges an incident, generates a local-model summary or sees the labeled fallback, and runs a workflow that records a simulated outcome.
- The UI remains useful when Wazuh, Ollama, or n8n is unavailable, and clearly reports the unavailable integration. No network response or external notification occurs.

5. Seven-day MVP delivery plan

Day 1 - Scope and local foundation

Confirm demo story, acceptance criteria, resource budget and n8n use case. Pin candidate component versions and verify their licenses. Set up isolated project, local network, sample schema and startup/reset scripts. Deliver: running skeleton and fictional asset catalog.

Day 2 - Synthetic event pipeline

Create deterministic scenarios and fixture replay, normalized incident API and source-mode indicator. Add seeded status and audit storage. Deliver: repeatable incident list and detail API, tested with reset and malformed fixtures.

Day 3 - Operator dashboard

Build overview, counts, filters, detail and evidence panels; implement acknowledgement and local notes. Deliver: usable synthetic-only dashboard with clear labels, empty/error states and no embedded credentials.

Day 4 - Local AI

Connect to a preinstalled local model; constrain incident context, prompt and output schema. Add timeout, verification label and prewritten fallback. Test missing model, slow response and prompt-injection text in a fixture. Deliver: honest, evidence-linked summaries.

Day 5 - Safe workflow

Create n8n webhook workflow that writes a simulated ticket or local log; add backend allowlist, validation, deduplication and visible execution status. Deliver: one end-to-end safe workflow and failure/retry demonstration.

Day 6 - Optional Wazuh lab and packaging

If machine resources and time permit, run the official single-node stack in isolation and ingest a test event using documented configuration. Implement a read-only adapter and verify the original alert. Otherwise keep the adapter interface and record this as unfinished. Freeze the demo, pin dependencies and document startup, reset, offline prerequisites and known limits.

Day 7 - Rehearsal and quality gate

Run clean-start and reset tests; rehearse all scenarios, unavailable integrations and fallback path. Review license inventory, visible labeling, presenter script and source attribution. Deliver a versioned demo package, quick-start guide, known-issues list and recorded acceptance results. Defer nonessential polish if a core scenario fails.

Critical path: synthetic event pipeline -> dashboard -> AI / workflow -> rehearsal. The Wazuh lab is valuable validation, but it must not be portrayed as complete if the adapter is not verified.

6. Practical sales demonstration

Before the meeting

- Start the isolated stack, confirm loopback-only access, pre-load the model and fixtures, and test a reset. Confirm no real data or external notification credentials are present.
- Check the AI fallback, n8n simulated output, timestamps and service health. Have a screenshot or offline recording ready in case the host is slow.

Suggested 10-15 minute walkthrough

0-2 min: Introduce the fictional ISP environment. Point to the synthetic-data and source-mode banner. State that Wazuh is the upstream security platform and the custom layer is the proposed service.

2-5 min: Filter the incident queue to a high-severity authentication scenario. Open the event and explain the asset, timestamps, original evidence and why it needs analyst attention.

5-8 min: Generate the local-model summary; compare one assertion to the evidence and identify uncertainty. Explain that the model is advisory and that the fallback is labeled.

8-11 min: Acknowledge the incident and run the simulated escalation. Show the local workflow result and activity history. Explain that no real ticket, alert or network change was sent.

11-15 min: Show a second scenario and the Wazuh lab source if actually connected. Discuss the customer discovery questions and next-step assessment, avoiding commitments on coverage, compliance or automatic protection.

Questions for a prospective customer

- What services and assets matter most: network infrastructure, servers, identity systems, endpoints or cloud? Which teams own them?
- Which log sources, formats, retention periods, alert volume, sites and operating constraints exist? What access and integration approvals are required?
- Where must data and models run? What are the security, availability, audit, incident response and licensing requirements? Who handles ongoing operations?

Recovery during a demo

If the model is slow, show the labeled fallback. If n8n is unavailable, show the failed state and explain the intended simulated outcome. If the Wazuh lab is down, switch to synthetic mode and explicitly say so. Reset the stack after each session; never imply an unavailable integration worked.

7. Isolation, security and operations

Demo controls

- Bind web interfaces to 127.0.0.1 by default; isolate containers on an internal network. Do not expose ports to the internet or customer networks. Restrict egress where practical, including after initial image and model downloads.
- Use unique local credentials, TLS where services cross hosts, secret files outside source control, minimal service permissions and no production credentials. Avoid privilege escalation or mounting the host Docker socket into application containers.
- Use fictional datasets, documentation IP addresses and fake identities. Review fixtures and screenshots before every presentation. Disable all external integrations and active response.
- Keep an audit record of demo acknowledgements and workflow attempts. Set a short retention period and provide an explicit reset that removes only demo-owned volumes and state.

Reliability and testing

Test API field mapping, filters and counts against known fixtures; verify source provenance; simulate missing services, timeouts, duplicate workflow calls and malformed AI output; inspect browser errors and accessibility basics. Measure first-start time, model latency and host resource use. Record the versions tested and known gaps.

Resource planning

Wazuh documentation lists at least 4 cores, 8 GB RAM and 50 GB disk for its single-node Docker stack alone. The custom services, n8n and local model need additional resources; a practical starting target is 16-32 GB RAM and 100 GB free disk, subject to measurement and model choice. A GPU is optional, but a slow CPU-only model can require the fallback. Offline use requires images, dependencies and model weights to be downloaded and tested in advance. No exact performance is promised.

Operational signals

Show integration connectivity, last successful refresh, model availability, workflow delivery status and fixture reset time. Record failures without dumping secrets or sensitive payloads. Pin image versions and model digest; document upgrade, restore and rollback procedures. Demo controls do not substitute for production authentication, backup, high availability, monitoring or incident response.

8. Future customer-specific deployment

A separate discovery and paid implementation should convert the demo into a customer-owned, customer-specific system. Prefer isolated deployments per customer until tenancy, security and commercial requirements justify an alternative.

Packaging and responsibilities

- Reusable core: versioned dashboard, backend, source adapters, workflow templates, schema, tests and deployment manifests. Keep each upstream component separate with version and license records.
- Customer overlay: asset groups, approved sources, branding, severity mapping, workflows, retention, integrations and environment-specific settings. Keep secrets in the customer-approved secret store, not in the package.
- Deployment: per-customer infrastructure, identity and access, TLS certificates, backup, logging, monitoring, network rules and change control. Establish ownership and boundaries in a statement of work.
- Acceptance: agree on supported sources and scenarios, service levels, data ownership, access controls, recovery targets and observable test cases before delivery. Test ingestion, alert fidelity, human approval, backup restore, upgrade and rollback in that environment.

Commercial path

Sell discovery, integration, customization, installation, support and training as clearly described services, subject to each component license. Quote only after sizing and defining scope. Select either a customer-owned n8n installation with the customer choosing its appropriate license, an appropriately licensed commercial offering, or a replaceable workflow engine. Do not assume the demo grants a redistribution or hosting right.

Dependencies and open decisions

Decide demo hardware and operating system; model and its license; exact Wazuh and n8n versions; fixture-only versus Wazuh lab scope; local persistence; which deployment ownership and service model to offer; whether customers may configure workflows; and support responsibilities. Before accepting a customer, verify actual data sources and obtain written authorization for access.

9. Licensing and attribution

This section is a planning checklist, not a legal opinion. Before distributing or selling a package, review the exact versions, licenses, dependencies and intended commercial arrangement with qualified counsel and the relevant vendors.

Wazuh

Wazuh states that its components use GPL version 2 and Apache License 2.0; individual components and dependencies have their own notices. Maintain a component inventory, source links, copyright and license notices. For any distribution, assess the applicable license obligations, including corresponding source and modification requirements where applicable. A separate service and API boundary does not automatically settle derivative-work questions. Preserve upstream branding and do not imply endorsement; review Wazuh trademark policy.

n8n

The self-hosted Community edition uses the Sustainable Use License, which is not a conventional open-source license. n8n's current guidance permits certain internal business uses and consulting, and says customer access to create or configure workflows, hosting n8n as a service, and white-labeling require a commercial license. Its guidance also describes installing and managing n8n on a client's own server, with the client choosing its plan. Confirm the exact delivery and interface design with n8n before sale; license rules can change.

AI model and other dependencies

Ollama software and each model and its weights have separate terms. Verify commercial use, redistribution, attribution, hosting, output and acceptable-use conditions for the selected model. Inventory frontend packages, base images, fonts and other dependencies, and deliver required notices and source offers as applicable. Do not bundle model weights until rights and artifact size are checked.

Release gate

Record component name, version, license, source, modification status, distribution method and required notices in a software bill of materials. Preserve the original licenses. Have a named owner resolve any unclear terms before packaging or a customer commitment.

10. References and implementation handoff

References checked 31 October 2026. Consult the current documentation again when selecting versions and before deployment.

Wazuh installation and component licenses
https://documentation.wazuh.com/current/installation-guide/index.html

Wazuh single-node Docker deployment and requirements
https://documentation.wazuh.com/current/deployment-options/docker/wazuh-container.html

Wazuh indexer API
https://documentation.wazuh.com/current/user-manual/indexer-api/index.html

Wazuh architecture
https://documentation.wazuh.com/current/getting-started/architecture.html

Wazuh trademark policy
https://wazuh.com/docs/legal-resources/Trademark_and_Brand_Policy_May_2024.pdf

n8n licensing use cases
https://support.n8n.io/article/can-i-use-your-license-for-my-use-case

Ollama generate API
https://docs.ollama.com/api/generate

Handoff checklist

- Approved scope and synthetic scenarios; versioned demo package; one-command start and scoped reset instructions; explicit source mode and fallback behavior.
- Architecture and interface documentation; tested service versions; presenter script; example outputs; health checks and known limitations.
- License and dependency inventory; exact customer deployment assumptions; acceptance results; list of open decisions and future work.

Next decision: approve or revise the proposed scope and hardware assumptions, then begin the seven-day build. The demo is intended to support a conversation and a tailored proposal, not to substitute for a production readiness assessment.